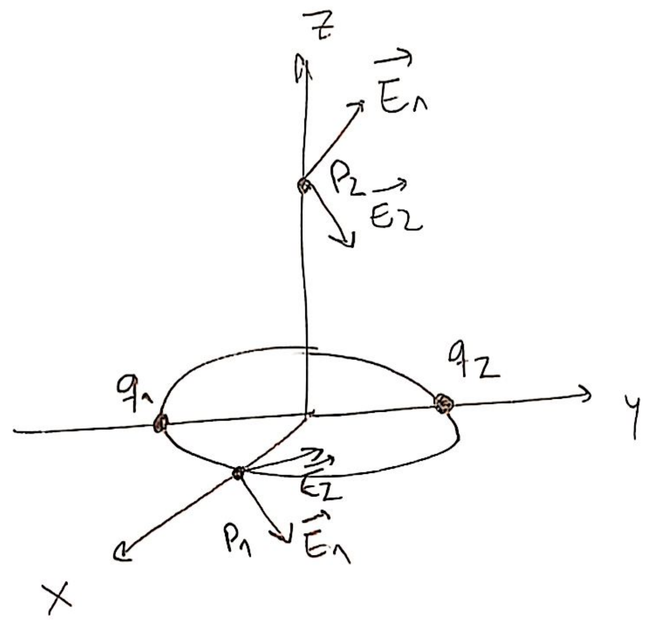

(1 pto cada una)

b) $\vec{E}(P_1) = \vec{E}_1 + \vec{E}_2$

$$\vec{E}_1 = \frac{k_e q_1 (\vec{r}_{P_1} - \vec{r}_1)}{|\vec{r}_{P_1} - \vec{r}_1|^3} = \frac{8{,}99 \cdot 10^9 \frac{Nm^2}{C^2} \cdot 10 \cdot 10^{-6} C \cdot (0{,}25\hat{\imath} + 0{,}25\hat{\jmath})\,m}{0{,}044\ m^3}$$

(1)
$$\begin{cases} \vec{r}_{P_1} = 0{,}25\,\hat{\imath}\ m \\ \vec{r}_1 = -0{,}25\,\hat{\jmath}\ m \\ \vec{r}_{P_1} - \vec{r}_1 = 0{,}25\hat{\imath} + 0{,}25\hat{\jmath}\ m \\ |\vec{r}_{P_1} - \vec{r}_1|^3 = \sqrt{(0{,}25)^2 + (0{,}25)^2}^{\,3} = 0{,}044\ m^3 \end{cases}$$

$\vec{E}_1 = 22475\hat{\imath} + 22475\hat{\jmath}\ \frac{N}{C}$ (1,5 ptos)

$$\vec{E}_2 = \frac{k_e q_2 (\vec{r}_{P_1} - \vec{r}_2)}{|\vec{r}_{P_1} - \vec{r}_2|^3} = \frac{8{,}99 \cdot 10^9 \frac{Nm^2}{C^2} \cdot (-5 \cdot 10^{-6} C)(0{,}25\hat{\imath} - 0{,}25\hat{\jmath})\ m}{0{,}044\ m^3}$$

1 pt
$$\begin{cases} \vec{r}_{P_1} = 0{,}25\,\hat{\imath}\ m \\ \vec{r}_2 = 0{,}25\,\hat{\jmath}\ m \\ \vec{r}_{P_1} - \vec{r}_2 = 0{,}25\hat{\imath} - 0{,}25\hat{\jmath}\ m \\ |\vec{r}_{P_1} - \vec{r}_2|^3 = \sqrt{(0{,}25)^2 + (0{,}25)^2}^{\,3} = 0{,}044\ m^3 \end{cases}$$

$\vec{E}_2 = -255397{,}7\hat{\imath} + 255397{,}7\hat{\jmath}\ \frac{N}{C}$

(1,5 ptos)

$\vec{E}_T = -232922{,}7\,\hat{\imath} + 277872{,}7\,\hat{\jmath}\ \frac{N}{C}$ (1 pt)

c) $\vec{E}(P_2) = \vec{E}_1 + \vec{E}_2$ (1 pto)

$$\vec{E}_1 = \frac{K_e q_1 (\vec{r}_{p2} - \vec{r}_1)}{|\vec{r}_{p2} - \vec{r}_1|^3} = \frac{8{,}99 \cdot 10^9 \frac{Nm^2}{C^2} \cdot 10 \cdot 10^{-6} C \cdot (0{,}25\hat{j} + 0{,}35\hat{k})m}{0{,}08\ m^3}$$

1,5 pts
$$\begin{cases} \vec{r}_{p2} = 0{,}35\hat{k}\ m \\ \vec{r}_1 = -0{,}25\hat{j}\ m \\ \vec{r}_{p2} - \vec{r}_1 = 0{,}25\hat{j} + 0{,}35\hat{k}\ m \\ |\vec{r}_{p2} - \vec{r}_1|^3 = \sqrt{(0{,}25)^2 + (0{,}35)^2}^3 = 0{,}08\ m^3 \end{cases}$$

$\vec{E}_1 = 280937{,}5\hat{j} + 393312{,}5\hat{k}\ \frac{N}{C}$

(1,5 pts)

$$\vec{E}_2 = \frac{K_e q_2 (\vec{r}_{p2} - \vec{r}_2)}{|\vec{r}_{p2} - \vec{r}_2|^3} = \frac{8{,}99 \cdot 10^9 \frac{Nm^2}{C^2} \cdot (-5 \cdot 10^{-6} C)(-0{,}25\hat{j} + 0{,}35\hat{k})m}{0{,}08\ m^3}$$

1,5 pts
$$\begin{cases} \vec{r}_{p2} = 0{,}35\hat{k}\ m \\ \vec{r}_2 = 0{,}25\hat{j}\ m \\ \vec{r}_{p2} - \vec{r}_2 = -0{,}25\hat{j} + 0{,}35\hat{k}\ m \\ |\vec{r}_{p2} - \vec{r}_2|^3 = \sqrt{(-0{,}25)^2 + (0{,}35)^2}^3 = 0{,}08\ m^3 \end{cases}$$

$\vec{E}_2 = 140468{,}75\hat{j} - 196656{,}25\hat{k}\ \frac{N}{C}$

(1,5 pts)

$$\vec{E}_T = 421406{,}25\hat{j} + 196656{,}25\hat{k}\ \frac{N}{C}$$

(1 pto)